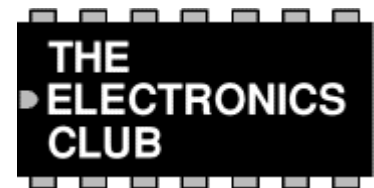


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Block Diagrams



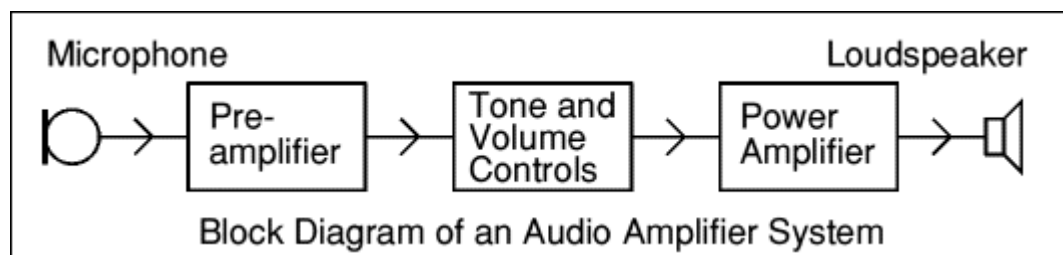
[Audio System](#) | [Radio System](#) | [Power Supply System](#) | [Feedback Control System](#)

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Block diagrams are used to understand (and design) complete circuits by breaking them down into smaller sections or blocks. Each block performs a particular function and the block diagram shows how they are connected together. No attempt is made to show the components used within a block, only the inputs and outputs are shown. This way of looking at circuits is called the **systems** approach.

Power supply (or battery) connections are usually not shown on block diagrams.

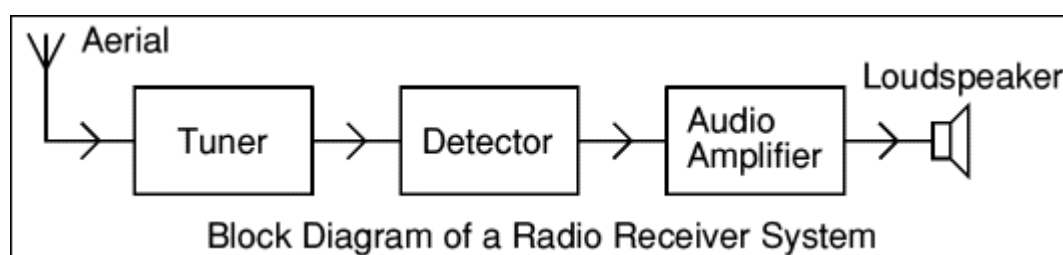
Audio Amplifier System



The power supply (not shown) is connected to the pre-amplifier and power amplifier blocks.

- **Microphone** - a transducer which converts sound to voltage.
- **Pre-Amplifier** - amplifies the small audio signal (voltage) from the microphone.
- **Tone and Volume Controls** - adjust the nature of the audio signal.
The tone control adjusts the balance of high and low frequencies.
The volume control adjusts the strength of the signal.
- **Power Amplifier** - increases the strength (power) of the audio signal.
- **Loudspeaker** - a transducer which converts the audio signal to sound.

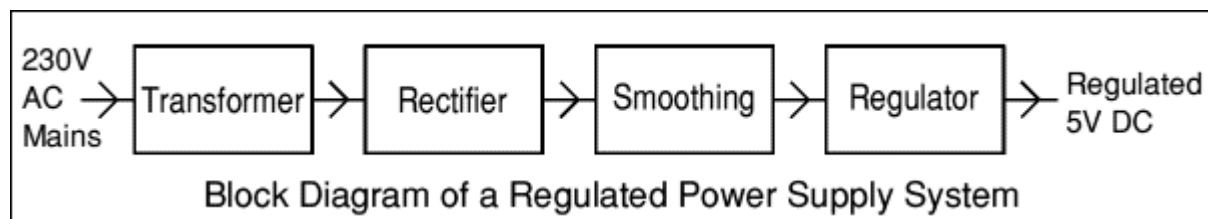
Radio Receiver System



The power supply (not shown) is connected to the audio amplifier block.

- **Aerial** - picks up radio signals from many stations.
- **Tuner** - selects the signal from just one radio station.
- **Detector** - extracts the audio signal carried by the radio signal.
- **Audio Amplifier** - increases the strength (power) of the audio signal.
This could be broken down into the blocks like the Audio Amplifier System shown above.
- **Loudspeaker** - a transducer which converts the audio signal to sound.

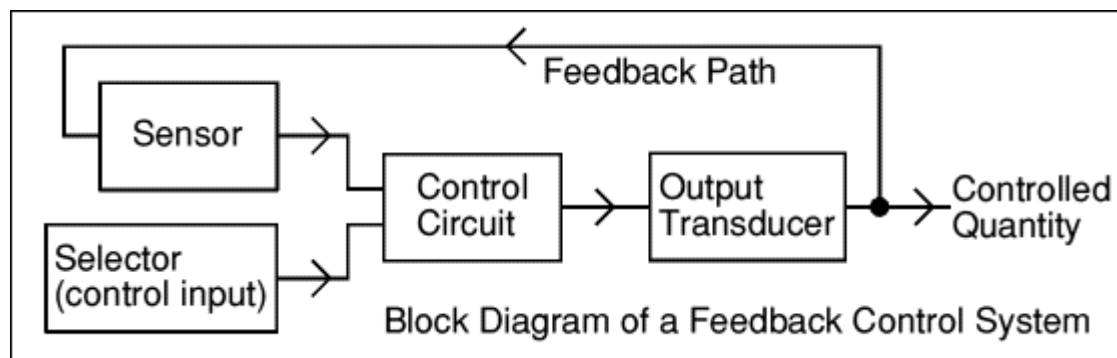
Regulated Power Supply System



- **Transformer** - steps down 230V AC mains to low voltage AC.
- **Rectifier** - converts AC to DC, but the DC output is varying.
- **Smoothing** - smooths the DC from varying greatly to a small ripple.
- **Regulator** - eliminates ripple by setting DC output to a fixed voltage.

For further information please see the [Power Supplies](#) page.

Feedback Control System



The power supply (not shown) is connected to the control circuit block.

- **Sensor** - a transducer which converts the state of the controlled quantity to an electrical signal.
- **Selector (control input)** - selects the desired state of the output. Usually it is a variable resistor.
- **Control Circuit** - compares the desired state (control input) with the actual state (sensor) of the controlled quantity and sends an appropriate signal to the output transducer.
- **Output Transducer** - converts the electrical signal to the controlled quantity.
- **Controlled Quantity** - usually not an electrical quantity, e.g. motor speed.
- **Feedback Path** - usually not electrical, the Sensor detects the state of the controlled

quantity.

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